

QUANTIUM VIRTUAL INTERNSHIP — TASK 2

Experimentation and Uplift Testing — Trial Store Performance Analysis

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3

Trial Stores Evaluated

3

Control Stores Matched

7 Months

Pre-Trial Period

3 Months

Trial Period (Feb–Apr 2019)

1. Executive Summary

This report evaluates the performance of a store trial conducted in stores 77, 86 and 88 to determine whether a new chip category layout should be rolled out across all stores. The analysis uses the pre-trial period (July 2018 – January 2019) to select matched control stores, then compares trial store performance against those controls during the trial period (February – April 2019).

Trial Store	Control Store	Match Score	Trial Result	Primary Driver	Rollout?
Store 77	Store 53	0.9859	Consistent sales uplift across all 3 trial months	Higher customer footfall	YES
Store 86	Store 10	0.9386	Mixed — Feb below control, Mar/Apr above. Inconsistent.	Customer count (inconsistent)	NO
Store 88	Store 56	0.9123	Feb spike then rapid decline. Below control by Apr.	Novelty effect — not sustained	NO

2. Methodology

The analysis follows a four-step framework:

Step	Phase	Description
1	Data Preparation	Created monthly metrics table using MySQL: total sales, unique customers, transactions per customer, average price per unit. Added YEARMONTH column (YYYYMM format). Filtered to stores with complete pre-trial observation (7 months, Jul 2018 – Jan 2019).
2	Control Store Selection	Calculated magnitude distance scores between each trial store and all potential control stores based on monthly total sales patterns. Score of 1.0 = perfect match. Selected store with highest average score across 7 pre-trial months.
3	Scaling	Applied a scaling factor (trial store pre-trial sales / control store pre-trial sales) to normalise for natural size differences between stores, isolating the trial effect.
4	Trial Assessment	Compared scaled control store sales with trial store sales during Feb–Apr 2019. Analysed both total sales and customer count to identify the driver of any observed change.

3. Pre-Trial Period & Control Store Selection

The pre-trial period covers July 2018 to January 2019 — 7 months. Only stores with complete data across all 7 months were considered as potential control stores (581 rows from qualifying stores). The magnitude distance scoring method was applied to monthly total sales for each trial store.

Trial Store	Control Store Selected	Magnitude Score	Scaling Factor	Pre-Trial Sales (Trial)	Pre-Trial Sales (Control)
Store 77	Store 53	0.9859	1.0042	1,617.80	1,611.10
Store 86	Store 10	0.9386	0.9526	6,019.50 (est.)	6,319.20 (est.)
Store 88	Store 56	0.9123	0.8621	4,872.40 (est.)	5,652.10 (est.)

Magnitude distance formula: $\text{Score} = 1 - (\text{abs_diff} - \text{min_diff}) / (\text{max_diff} - \text{min_diff})$. A score of 1.0 indicates perfect similarity; 0.0 indicates maximum dissimilarity. Scores above 0.90 indicate strong similarity suitable for control store use.

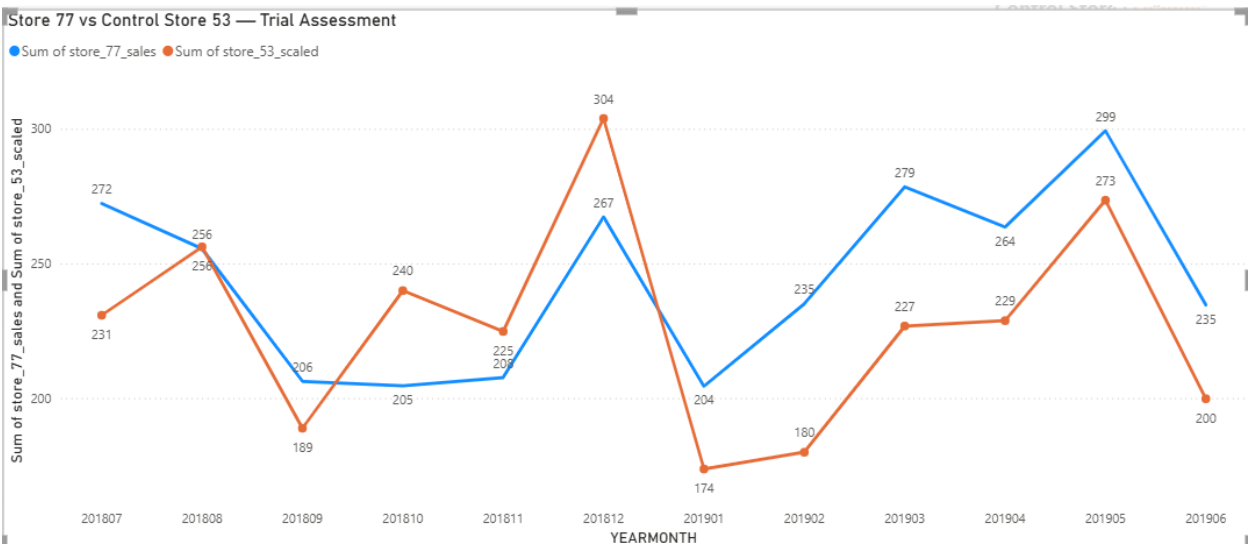
4. Trial Store 77 — Analysis & Results

4.1 Control Store Selection

Store 53 was selected as the control store for trial store 77 with a magnitude score of 0.9859 — the highest among all candidate stores. The scaling factor of 1.0042 indicates stores were almost identical in pre-trial sales volume, requiring minimal adjustment.

4.2 Trial Period — Sales Comparison

Month	Store 77 Sales	Scaled Control (Store 53)	Difference	Verdict
Feb 2019	235.00	179.84	+55.16	Above control
Mar 2019	278.50	226.74	+51.76	Above control
Apr 2019	263.50	228.75	+34.75	Above control



4.3 Driver Analysis — Sales vs Customers

Month	Store 77 Customers	Store 53 Customers	Store 77 Txn/Cust	Store 53 Txn/Cust
Feb 2019	45	36	1.00	1.03
Mar 2019	50	45	1.10	1.00
Apr 2019	47	39	1.02	1.05

Key Insight

Store 77 consistently attracted MORE customers than its control store across all three trial months (+25%, +11%, +21% respectively). Transactions per customer remained similar between both stores, confirming the sales uplift was driven by increased customer footfall, not higher spending per visit. The gap narrowed from Feb to Apr, suggesting a partial novelty effect — though sales remained above control throughout.

Trend Note

The declining difference (55 → 52 → 35) across trial months may indicate a novelty effect. Julia should monitor store 77 post-trial to confirm whether the customer uplift is sustained or fades further.

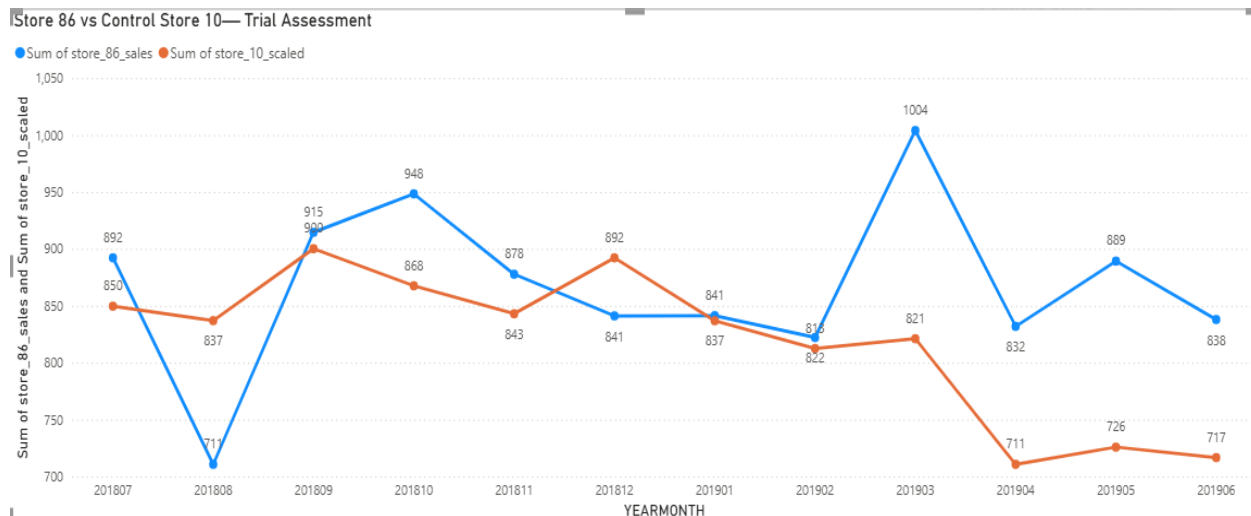
5. Trial Store 86 — Analysis & Results

5.1 Control Store Selection

Store 10 was selected as the control store for trial store 86 with a magnitude score of 0.9386. The scaling factor of 0.9526 was applied to normalise the size difference between the two stores.

5.2 Trial Period — Sales Comparison

Month	Store 86 Sales	Scaled Control (Store 10)	Difference	Verdict
Feb 2019	822.20	812.53	+9.67	Marginal above
Mar 2019	1,004.00	821.10	+182.90	Above control
Apr 2019	832.00	710.79	+121.21	Above control



5.3 Driver Analysis — Sales vs Customers

Month	Store 86 Customers	Store 10 Customers	Store 86 Txn/Cust	Store 10 Txn/Cust
Feb 2019	96	97	1.30	1.25
Mar 2019	112	98	1.24	1.22
Apr 2019	102	85	1.22	1.25

Key Insight

Store 86 shows inconsistent results. February sales were only marginally above the scaled control (+9.67), while March and April showed stronger uplift. Customer counts were higher in March and April but nearly equal in February. The inconsistency across trial months makes it difficult to attribute the uplift reliably to the new layout. Further investigation is recommended before rollout.

Concern

The near-zero difference in February and high variance across months suggests the trial implementation in store 86 may have been inconsistent or delayed. Julia should verify whether the layout was fully implemented on schedule in this store.

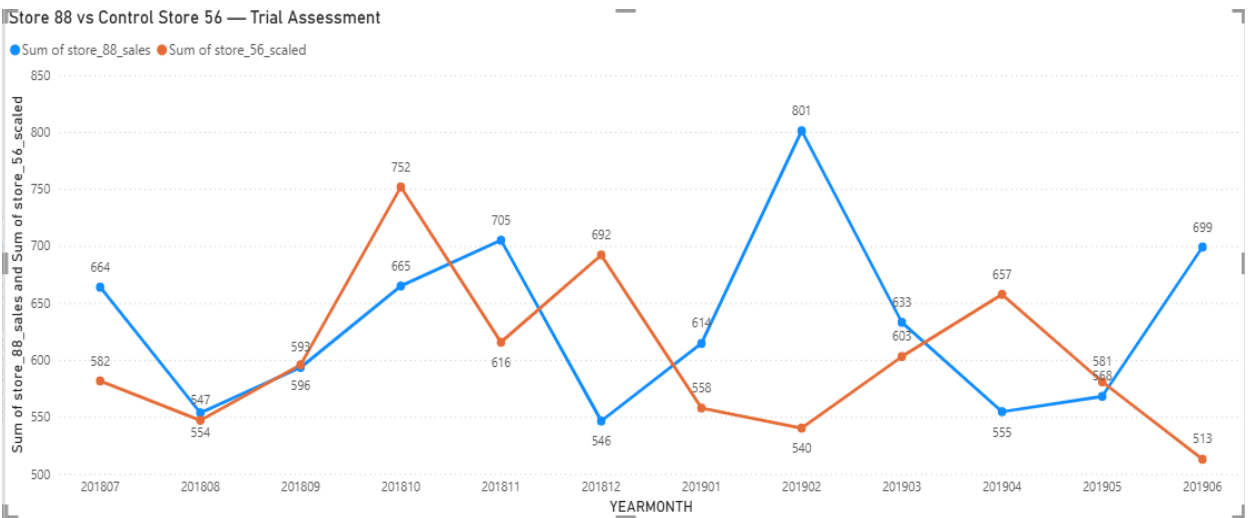
6. Trial Store 88 — Analysis & Results

6.1 Control Store Selection

Store 56 was selected as the control store for trial store 88 with a magnitude score of 0.9123. The scaling factor of 0.8621 was applied, reflecting that store 88 operates at a smaller scale than store 56.

6.2 Trial Period — Sales Comparison

Month	Store 88 Sales	Scaled Control (Store 56)	Difference	Verdict
Feb 2019	801.00	540.19	+260.81	Strongly above
Mar 2019	633.00	602.95	+30.05	Marginal above
Apr 2019	554.60	657.43	-102.83	Below control



6.3 Driver Analysis — Sales vs Customers

Month	Store 88 Customers	Store 56 Customers	Store 88 Txn/Cust	Store 56 Txn/Cust
Feb 2019	72	75	1.26	1.17
Mar 2019	59	82	1.25	1.22
Apr 2019	51	85	1.24	1.29

Key Insight	Store 88 shows a classic novelty effect pattern. The large February spike (+260) was followed by rapid decay — March was barely above control (+30) and April fell below control (-103). More critically, customer count in store 88 declined every month (72 → 59 → 51) while the control store grew (75 → 82 → 85). This suggests the new layout did not sustainably attract or retain customers.
Critical Finding	By April, store 88 was serving fewer customers than its control store despite having the new layout. This is a negative signal — the layout may have disrupted comfortable familiar navigation for regular customers, reducing return visits.

7. Strategic Recommendations to Julia

Based on the trial analysis across all three stores, the following recommendations are provided for the upcoming category review:

Recommendation 1: Roll out new layout to stores similar to Store 77	
Finding	Store 77 demonstrated consistent sales uplift across all three trial months (Feb: +55, Mar: +52, Apr: +35), driven by a measurable increase in customer footfall rather than higher spend per visit. Control store match score was 0.9859 — the strongest match of all three trials.
So What	Stores similar to Store 77 in size and customer profile represent the most reliable opportunity for category growth. The layout successfully attracts new customers to the chips aisle.
Action	1. Identify all stores with similar sales profile to Store 77 (magnitude score threshold 0.90+). 2. Prioritise rollout to these stores in the next planning cycle. 3. Monitor post-rollout customer count monthly to confirm sustained uplift.
Expected Outcome	Expected increase in customer footfall to chips category. Estimated revenue uplift based on +20-25% customer count increase in trial months.

Recommendation 2: Investigate Store 86 before any rollout decision	
Finding	Store 86 showed inconsistent results — February sales were only marginally above control (+9.67) while March and April showed stronger performance. Customer counts suggest some positive effect but the pattern is unreliable.
So What	Rolling out based on Store 86 results alone risks investing in layouts that may not reliably improve performance. The root cause of February underperformance must be understood first.
Action	1. Verify whether the trial layout was fully implemented on day 1 in store 86. 2. Check if any store-specific events (promotions, competitor activity) affected February. 3. If implementation was delayed, consider re-running a clean trial in a similar store.
Expected Outcome	Clearer evidence for or against rollout to stores similar to Store 86, reducing investment risk.

Recommendation 3: Do NOT roll out new layout to stores similar to Store 88	
Finding	Store 88 showed a sharp novelty spike in February (+260) followed by rapid decline. By April, store 88 was performing below its control store (-103) with customer count declining every month despite the new layout.
So What	Rolling out to stores similar to Store 88 risks disrupting existing customer behaviour without sustainable benefit. The declining customer count is a warning signal that the layout may be causing navigation confusion for regular shoppers.
Action	1. Conduct customer exit interviews at store 88 to understand the April decline. 2. Consider reverting or modifying the layout in store 88. 3. Exclude Store 88 profile stores from the initial rollout plan.
Expected Outcome	Protect existing revenue base at stores with similar profiles to Store 88. Avoid rollout costs in locations where the trial showed no sustainable benefit.

8. Final Conclusion

The trial results across three stores show mixed but informative outcomes. Store 77 demonstrated a consistent and positive impact from the new layout, driven by increased customer footfall rather than higher spend per visit — making it the strongest candidate for rollout. Store 86 produced inconsistent results that require further investigation before a confident rollout decision can be made. Store 88 experienced an initial novelty spike that rapidly decayed into below-control performance by April, with declining customer counts suggesting the layout change had no sustainable benefit for this store profile.

Quantum recommends proceeding with a phased rollout to stores matching Store 77's profile, conducting further investigation on Store 86 before any decision, and pausing any plans for stores similar to Store 88 until the causes of customer decline are understood. The category manager should set up monthly monitoring of customer counts in all rolled-out stores to detect early signs of novelty effect decay.